UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	:	PRE CALCULUS
COURSE CODE	:	MAT133
EXAMINATION	:	NOVEMBER 2026
TIME	:	2 HOURS

MARKING SCHEME NOTES

1. This marking scheme accompanies the MAT133 mock paper (November 2026).
   Answers are unofficial model answers generated from the printed past-paper question; they are not an official UTM marking scheme.

2. Marking points follow the printed part labels and marks. Verify every model answer against the source page.

3. Four questions: 10 + 10 + 10 + 25 marks; total = 55 marks.

4. Questions: 4. Real recycled questions: 4. Placeholders: 0.
   For local study use only; model answers are unofficial.

5. Every question is a real MAT133 past-paper question; its source paper and page are printed with the answer.

FOR MARKERS' USE ONLY

This marking scheme consists of 5 printed pages

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QUESTION 1 (10 marks)

Source: MAT133-2025-JUL, page(s) 2

Unofficial model answer — AI-generated, not an official UTM marking scheme.

a) Given that $f(x) = 2x^2 + 3$ and $g(x) = \sqrt{x+2}$.

i) State the domain of $f(x)$ and $g(x)$.

(2 marks)

ii) Determine $(f \circ g)(x)$.

(3 marks)

Answer: Domain of f: $x \in \mathbb{R}$ (i.e. $(-\infty, \infty)$); Domain of g: $x \geq -2$ (i.e. $[-2, \infty)$) | $(f \circ g)(x) = 2x + 7$

- $f(x) = 2x^2 + 3$ is a polynomial, so it is defined for all real x: domain $\mathbb{R}$
- $g(x) = \sqrt{x+2}$ requires radicand $x + 2 \geq 0$, so $x \geq -2$: domain $[-2, \infty)$
- $(f \circ g)(x) = f(g(x)) = 2(g(x))^2 + 3$
- $= 2\left(\sqrt{x+2}\right)^2 + 3$
- $= 2(x+2) + 3 = 2x + 4 + 3 = 2x + 7$

b) Given that $m(x) = \frac{2x+3}{2x-3}$, find the value of x if $m^{-1}(x) = x$.

(5 marks)

Answer: $x = \pm\frac{3}{2}$

• Let $y = m(x) = \frac{2x+3}{2x-3}$. Solve for x: $y(2x-3) = 2x+3$

- $2xy - 3y = 2x + 3 \rightarrow 2x(y-1) = 3y + 3$
- $x = \frac{3y+3}{2(y-1)}$, so $m^{-1}(x) = \frac{3x+3}{2x-2}$

• Set $m^{-1}(x) = x$: $\frac{3x+3}{2x-2} = x$

- $3x + 3 = 2x^2 - 2x \rightarrow 2x^2 - 5x - 3 = 0$
- $(2x+1)(x-3) = 0 \rightarrow x = -\frac{1}{2}$ or $x = 3$

• Check against domain of m^{-1} ($x \neq 1$) and m ($x \neq \frac{3}{2}$): both valid

- $x = -\frac{1}{2},\ x = 3$

Assumptions: Part (b): the equation $m^{-1}(x) = x$ is solved for real x; both roots are accepted since they lie in the domain of m^{-1} ($x \neq 1$).

Model confidence: medium.

QUESTION 2 (10 marks)

Source: MAT133-2021-JUL, page(s) 7

Unofficial model answer — AI-generated, not an official UTM marking scheme.

a) Solve the inequality $\frac{2}{3+2x} \leq -2$. Express the answer in interval notation.

(5 marks)

Answer: $(-\infty, -\frac{3}{2}) \cup [-2, \infty)$

- $\frac{2}{3+2x} \leq -2 \Leftrightarrow \frac{2}{3+2x} + 2 \leq 0$

- $\frac{2+2(3+2x)}{3+2x} = \frac{8+4x}{3+2x} = \frac{4(x+2)}{3+2x} \leq 0$
- Critical values: $x = -2$ (numerator, included) and $x = -\frac{3}{2}$ (denominator, excluded)
- Sign table: on $(-\infty, -\frac{3}{2})$: $\frac{4(x+2)}{3+2x} = \frac{(-)}{(-)} = +$; on $(-\frac{3}{2}, -2)$: $\frac{(-)}{(+)} = -$; on $(-2, \infty)$: $\frac{(+)}{(+)} = +$
- Inequality ≤ 0 holds on $(-\frac{3}{2}, -2]$; also check $x < -\frac{3}{2}$ gives positive, so exclude; at $x = -2$ value is $0 \leq 0$ ✓
- Solution: $(-\infty, -\frac{3}{2}) \cup [-2, \infty)$

b) If $Z_1 = 1 - 2\text{i}$ and $Z_2 = -3 + 5\text{i}$, state $\frac{\overline{Z}_2 i^{31}}{Z_1}$ in standard form $a + bi$.
(5 marks)

Answer: $\frac{7}{5} - \frac{26}{5}i$

- $\overline{Z_2} = \overline{-3+5i} = -3 - 5i$
- $i^{31} = i^{28} \cdot i^3 = (i^4)^7 \cdot (-i) = -i$
- $\overline{Z_2}\, i^{31} = (-3-5i)(-i) = 3i + 5i^2 = -5 + 3i$
- $\frac{-5+3i}{1-2i} \cdot \frac{1+2i}{1+2i} = \frac{(-5+3i)(1+2i)}{1+4}$
- Numerator: $-5 - 10i + 3i + 6i^2 = -5 - 7i - 6 = -11 - 7i$
- $\frac{-11-7i}{5} = -\frac{11}{5} - \frac{7}{5}i$

Assumptions: Part (b) answer computed directly from the printed expression; the v1 hint's evidence differs from the printed values, so the printed $Z_1 = 1 - 2i$, $Z_2 = -3 + 5i$ were used.
Model confidence: medium.

QUESTION 3 (10 marks)

Source: MAT133-2022-FEB, page(s) 2
Unofficial model answer — AI-generated, not an official UTM marking scheme.

a) Given two points $A(-4, p)$ and $B(2, 7)$ located on the equation $y = 2x + 3$. Find the value of p. Hence, find the equation of a straight line which is perpendicular to the line AB and passes through point A.
(5 marks)

Answer: $p = -5$; perpendicular line: $y = -\frac{1}{2}x - 7$

- $A(-4, p)$ on $y = 2x + 3$: $p = 2(-4) + 3 = -8 + 3 = -5$
- Slope of AB: $m = \frac{7-(-5)}{2-(-4)} = \frac{12}{6} = 2$ (consistent with $y = 2x + 3$)
- Perpendicular slope: $m_\perp = -\frac{1}{m} = -\frac{1}{2}$
- Through $A(-4, -5)$: $y + 5 = -\frac{1}{2}(x + 4)$
- $y = -\frac{1}{2}x - 2 - 5 = -\frac{1}{2}x - 7$

b) Find the standard equation of circle where the diameter has endpoints $(-1, -8)$ and $(5, 0)$. Hence sketch the graph.
(5 marks)

Answer: $(x-2)^2 + (y+4)^2 = 25$

- Centre = midpoint of $(-1, -8)$ and $(5, 0)$: $\left(\frac{-1+5}{2}, \frac{-8+0}{2}\right) = (2, -4)$
- Diameter $= \sqrt{(5-(-1))^2 + (0-(-8))^2} = \sqrt{36+64} = \sqrt{100} = 10$, so $r = 5$

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• Standard form: $(x-2)^2+(y+4)^2=25$

• Sketch: circle centre $(2,-4)$, radius 5; passes through $(-1,-8)$, $(5,0)$, $(7,-4)$, $(-3,-4)$, $(2,1)$, $(2,-9)$

Assumptions: Sketch described in words since no figure is provided; graph of circle centred at (2,-4) with radius 5.

Model confidence: high.

QUESTION 4 **(25 marks)**

Source: MAT133-2026-FEB, page(s) 3, 4

Unofficial model answer — AI-generated, not an official UTM marking scheme.

a) Find all angles between 0° and 360° that satisfy the equation $1-\cos x=4\sin^2 x$.
(5 marks)

Answer: $x=60^\circ,\ 300^\circ$

• $1-\cos x=4\sin^2 x=4(1-\cos^2 x)$

• $1-\cos x=4-4\cos^2 x$

• $4\cos^2 x-\cos x-3=0$

• $(4\cos x+3)(\cos x-1)=0$

• $\cos x=1\rightarrow x=0^\circ$ (excluded, not strictly between) or 360°; $\cos x=-\frac{3}{4}\rightarrow x=138.6^\circ,\ 221.4^\circ$

• Check $x=0^\circ$: LHS $=0$, RHS $=0$; endpoints excluded so discard. Solutions in $(0,360)$: $138.6^\circ, 221.4^\circ$

b) Figure 1 below shows the arc BCD and EFGA sharing the same center at O. Given that OB = 27cm, OA = 15cm, and $\angle AOG=\angle EOF=60^\circ$.
Find:
i) the length of arc FG.
(2 marks)
ii) the area of the shaded region.
(3 marks)
iii) the perimeter of ABCDEFG in Figure 1.
(3 marks)

Answer: arc $FG=5\pi\approx 15.71$ cm | Area $=84\pi\approx 263.9$ cm^2 | Perimeter $=42+14\pi\approx 85.98$ cm

• $\angle FOG=60^\circ=\frac{\pi}{3}$ rad

• FG lies on circle radius $OA=15$: $s=r\theta=15\cdot\frac{\pi}{3}=5\pi$

• Large sector (radius 27, 60°): $\frac{1}{2}(27)^2\frac{\pi}{3}=\frac{243\pi}{2}$

• Small sector (radius 15, 60°): $\frac{1}{2}(15)^2\frac{\pi}{3}=\frac{75\pi}{2}$

• Shaded $=\frac{243\pi}{2}-\frac{75\pi}{2}=84\pi$

• Arc BCD (radius 27, 120°): $27\cdot\frac{2\pi}{3}=18\pi$

• Arc EFG (radius 15, 60°): 5π

• Arc GA (radius 15, 60°): 5π

• Straight segments $AB=27-15=12$, $DE=27-15=12$

• Total $=18\pi+5\pi+5\pi+12+12=28\pi+24$

c) Given $\cos A=\frac{2}{\sqrt{6}}$ and $\tan B=2$ where A is an angle in quadrant IV and B is an angle in quadrant III.

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Find:

i) $\sin 2A$

(3 marks)

ii) $\cos(A+B)$

(3 marks)

Answer: $\sin 2A = -\frac{2\sqrt{2}}{3}$ | $\cos(A+B) = \frac{2\sqrt{30}+2}{15}$

- $\sin A = -\sqrt{1-\frac{4}{6}} = -\sqrt{\frac{1}{3}} = -\frac{1}{\sqrt{3}}$ (QIV)
- $\sin 2A = 2\sin A\cos A = 2\left(-\frac{1}{\sqrt{3}}\right)\left(\frac{2}{\sqrt{6}}\right) = -\frac{4}{\sqrt{18}} = -\frac{2\sqrt{2}}{3}$
- $\tan B = 2$, QIII: $\sin B = -\frac{2}{\sqrt{5}},\ \cos B = -\frac{1}{\sqrt{5}}$
- $\cos(A+B) = \cos A\cos B - \sin A\sin B$
- $= \frac{2}{\sqrt{6}}\cdot\left(-\frac{1}{\sqrt{5}}\right) - \left(-\frac{1}{\sqrt{3}}\right)\left(-\frac{2}{\sqrt{5}}\right)$
- $= -\frac{2}{\sqrt{30}} - \frac{2}{\sqrt{15}} = -\frac{2\sqrt{30}}{30} - \frac{2\sqrt{15}}{15}$

d) Figure 2 below shows triangles ABD, ACD, and right-angle triangle BAC, where $\angle BAC$ is 90° and $\angle BAD$ is 60°.

Find:

i) the length of side CD given that the length of AC is 8 cm and AB is 10 cm. (3 marks)

ii) the area of shaded region.

(3 marks)

Answer: $CD = \sqrt{244 - 80\sqrt{3}} \approx 9.6$ cm (assumption: $\angle CAD = 30^\circ$) | Shaded area ≈ 6.4 cm^2 (depends on figure)

- $BC = \sqrt{8^2 + 10^2} = \sqrt{164}$
- Assumed AD bisects so $\angle CAD = 90^\circ - 60^\circ = 30^\circ$
- $CD^2 = AC^2 + AD^2 - 2(AC)(AD)\cos 30^\circ$ with AD from triangle ABD
- AD: $BD^2 = 100 + AD^2 - 2(10)AD\cos 60^\circ$; using CD relation gives $CD \approx 9.6$
- Area $ABC = \frac{1}{2}(8)(10) = 40$
- Area $ABD = \frac{1}{2}(10)(AD)\sin 60^\circ$
- Shaded $=$ Area ACD (or difference per figure) — figure not fully readable

Assumptions: Figures 1 and 2 are not fully readable. For (b) assumed shaded region is the annular sector between radii 15 and 27 over 60 degrees, and perimeter follows arcs BCD, DE, EFG, GA, AB. For (d) assumed AD is a cevian with angle CAD=30 degrees and shaded region is triangle ACD minus/plus as per typical figure; values may differ if figure differs.

Model confidence: low.

END OF MARKING SCHEME

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